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# COMPLETE DATA ANALYSIS WITH MISSING DATA, THE MCAR TEST, IMPUTATION, LOF OUTLIERS

# ------------------------ 0. SETUP ---------------------------------------

library(dplyr)

library(visdat)

library(naniar)

library(mice)

library(ggplot2)

library(tidyr)

library(dbscan)


# File path (adjust as needed)

data_path <- "C:/Users/pc/Desktop/TANKOFOLDER/DataSet_No_Details.csv"


# ------------------------ 1. READ & CLEAN DATA ---------------------------

df <- read.csv(data_path)


# Remove columns not needed

cols_to_remove <- c("h_index_34", "h_index_56", "hormone10_1", "hormone10_2",

              "an_index_23", "outcome", "factor_eth", "factor_h",

              "factor_pcos", "factor_prl")

MD_df <- df %>% select(-any_of(cols_to_remove))


# Keep factor variables separately (optional)

factor_df <- df %>% select(record_id, outcome, factor_eth, factor_h,

                  factor_pcos, factor_prl)
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# ------------------------ 2. MISSING DATA ANALYSIS ----------------------

total_na <- sum(is.na(MD_df))

na_per_col <- colSums(is.na(MD_df))

na_percent <- colMeans(is.na(MD_df)) * 100


cat("\n=== Missing Data Overview ===\n")

cat("Total NAs:", total_na, "\n")

print(na_per_col[na_per_col > 0])


# Columns with >35% missing – will be removed

high_miss <- names(na_percent[na_percent > 35])

cat("\nColumns with >35% missing:\n"); print(high_miss)


# Visualise missing patterns

vis_miss(MD_df)

gg_miss_var(MD_df)


# Remove high-   missing columns

cols_to_remove1 <- c("hormone9", "hormone11", "hormone12", "hormone13", "hormone14")

handle_MD_df <- MD_df %>% select(-any_of(cols_to_remove1))


# ------------------------ 3. LITTLE'S MCAR TEST (HOMEWORK) ----------------

numeric_data <- handle_MD_df %>% select(where(is.numeric))

mcar_result <- mcar_test(numeric_data)
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p_value <- mcar_result$p.value

cat("\n=== Little's MCAR Test ===\n")
print(mcar_result)
cat("\nInterpretation:\n")
if (p_value > 0.05) {
  cat("p =", p_value, "> 0.05 → Data is MCAR. Multiple imputation is suitable.\n")
} else {
  cat("p =", p_value, "≤ 0.05 → Data is NOT MCAR. Use advanced methods.\n")
}

# ------------------------ 4. MULTIPLE IMPUTATION (PMM) -------------------
imputed_object <- mice(handle_MD_df, m = 5, method = "pmm", print = FALSE)
imputed_full <- complete(imputed_object)

# Remove any rows with remaining NAs (required for LOF)
imputed_clean <- na.omit(imputed_full)
cat("\nRows removed due to NAs:", nrow(imputed_full) - nrow(imputed_clean), "\n")

# ------------------------ 5. DENSITY PLOT (ORIGINAL vs IMPUTED) ----------
if ("hormone10_generated" %in% names(handle_MD_df)) {
  ggplot() +
    geom_density(data = handle_MD_df, aes(x = hormone10_generated, fill = "Original"),
                 alpha = 0.5, na.rm = TRUE) +
    geom_density(data = imputed_clean, aes(x = hormone10_generated, fill = "Imputed"),
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        alpha = 0.5, na.rm = TRUE) +
    labs(title = "Density: Original vs Imputed (hormone10_generated)") +
    scale_x_continuous(limits = c(0,2)) +
    theme_minimal()
} else {
  first_num <- names(handle_MD_df)[sapply(handle_MD_df, is.numeric)][1]
  rbind(data.frame(value = handle_MD_df[[first_num]], Method = "Original"),
       data.frame(value = imputed_clean[[first_num]], Method = "Imputed")) %>%
    ggplot(aes(x = value, fill = Method)) +
    geom_density(alpha = 0.5, na.rm = TRUE) +
    labs(title = paste("Density:", first_num, "- Original vs Imputed")) +
    theme_minimal()
}


# ------------------------ 6. TRADITIONAL OUTLIER BOXPLOTS ----------------
# Lipid variables
lipid_cols <- c("lipids1", "lipids2", "lipids3", "lipids4", "lipids5")
lipid_present <- lipid_cols[lipid_cols %in% names(imputed_clean)]
if (length(lipid_present) > 0) {
  imputed_clean %>%
    select(all_of(lipid_present)) %>%
    pivot_longer(everything(), names_to = "var", values_to = "val") %>%
    ggplot(aes(x = var, y = val)) +
    geom_boxplot(fill = "lightblue", alpha = 0.7) +
    labs(title = "Outlier Detection - Lipid Variables") +
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  theme_minimal()
}

# All numeric variables
imputed_clean %>%
 select(where(is.numeric)) %>%
 pivot_longer(everything(), names_to = "var", values_to = "val") %>%
 ggplot(aes(y = val)) +
 geom_boxplot(fill = "steelblue", alpha = 0.6, outlier.color = "red") +
 facet_wrap(~var, scales = "free", ncol = 4) +
 labs(title = "Boxplots for All Numeric Variables") +
 theme_minimal() +
 theme(axis.text.x = element_blank())

# ------------------------ 7. LOCAL OUTLIER FACTOR (LOF) ------------------
lof_dataset <- imputed_clean
lof_data <- lof_dataset %>% select(where(is.numeric))

# Remove constant columns
constant <- sapply(lof_data, function(x) sd(x, na.rm = TRUE) == 0)
if (any(constant)) lof_data <- lof_data %>% select(-which(constant))

# Scale
lof_scaled <- scale(lof_data)
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# LOF parameters
minPts_val <- min(20, floor(nrow(lof_scaled) / 5))
cat("\nLOF using minPts =", minPts_val, "\n")

# Compute LOF
lof_scores <- lof(lof_scaled, minPts = minPts_val)

# Add scores and outlier flag
lof_dataset$LOF_Score <- lof_scores
threshold <- 1.5
lof_dataset$Is_Outlier <- lof_scores > threshold

# Summary
cat("\nLOF Score Statistics:\n")
print(summary(lof_scores))
cat("\nOutliers detected:", sum(lof_dataset$Is_Outlier),
    "(", round(mean(lof_dataset$Is_Outlier)*100, 2), "%)\n")

# Top 10 outliers
cat("\nTop 10 outliers by LOF score:\n")
lof_dataset %>%
  arrange(desc(LOF_Score)) %>%
  select(where(is.numeric), LOF_Score, Is_Outlier) %>%
  head(10) %>%
  print()
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# ------------------------- 8. LOF VISUALISATIONS --------------------------

# Histogram

ggplot(lof_dataset, aes(x = LOF_Score)) +

  geom_histogram(aes(fill = after_stat(count)), bins = 40, color = "black") +

  geom_vline(xintercept = threshold, color = "red", linetype = "dashed", size = 1) +

  geom_vline(xintercept = 1, color = "blue", linetype = "dotted") +

  scale_fill_gradient(low = "lightblue", high = "darkblue") +

  labs(title = "LOF Distribution",

       subtitle = paste("minPts =", minPts_val, "|", sum(lof_dataset$Is_Outlier), "outliers")) +

  theme_minimal()


# Bivariate scatterplot (first two numeric variables)

if (ncol(lof_data) >= 2) {

  var1 <- names(lof_data)[1]

  var2 <- names(lof_data)[2]

  ggplot(lof_dataset, aes(x = .data[[var1]], y = .data[[var2]])) +

    geom_point(aes(color = Is_Outlier, size = LOF_Score), alpha = 0.6) +

    scale_color_manual(values = c("FALSE" = "blue", "TRUE" = "red")) +

    labs(title = "Bivariate Scatterplot with LOF Outliers") +

    theme_minimal()

}


# Boxplot of LOF scores by outlier status

ggplot(lof_dataset, aes(x = Is_Outlier, y = LOF_Score, fill = Is_Outlier)) +
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  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = threshold, linetype = "dashed", color = "darkred") +
  scale_fill_manual(values = c("FALSE" = "lightblue", "TRUE" = "red")) +
  labs(title = "LOF Score: Normal vs Outlier") +
  theme_minimal()


# ------------------------ 9. FINAL SUMMARY ------------------------------
cat("\n========== FINAL SUMMARY ==========\n")
cat("1. Little's MCAR Test: p =", p_value,
    ifelse(p_value > 0.05, "→ Data is MCAR", "→ Data is NOT MCAR"), "\n")
cat("2. Imputation: PMM, original NAs =", sum(is.na(handle_MD_df)),
    ", after cleaning =", sum(is.na(imputed_clean)), "\n")
cat("3. LOF: minPts =", minPts_val, ", outliers =", sum(lof_dataset$Is_Outlier),
    "(", round(mean(lof_dataset$Is_Outlier)*100, 2), "%)\n")
cat("4. LOF score range:", round(min(lof_scores), 3), "–", round(max(lof_scores), 3), "\n")
cat("\nAnalysis completed successfully.\n")
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